

Hanoi Urban Noise Mapping

Data Collection Report

363 smartphone measurements · 3 sites · 10 Jun to 22 Jul 2026

363	3	47-88	0.45	39%
Measurements	Sites	dB range	Model R ²	Exceed QCVN

1. Objective

Map and characterise urban noise in three contrasting districts of Hanoi, and predict noise from urban morphology. We follow the Sunbird AI methodology (Urban Noise Uganda 61K, Nsumba et al., 2026): smartphone field collection plus a morphology-to-noise model, reproduced then applied here.

2. Data collection summary

Site	n	Median dB	Min-Max	% < 60 dB	% roadside
Ocean Park	184	68	47-88	28%	73%
Hoan Kiem lake	99	70	55-80	7%	89%
Vinh Tuy area	80	66	53-76	20%	74%
ALL	363	68	47-88	21%	78%

Zones: Ocean Park = new high-rise development (shielding, construction); Vinh Tuy = transport infrastructure (heavy traffic); Hoan Kiem = historic old quarter (narrow streets, pedestrian zones). QC: site labels cross-checked against GPS (6 mislabelled submissions reassigned).

Headline findings: median 68 dB, 39% of measurements exceed the QCVN limit, noise peaks at 18:00. A model trained directly on our data reaches R² 0.45 (honest cross-validation, section 5).

3. Methodology

3.1 Field data-collection protocol

Study zones were mapped in Google Earth and chosen for contrasting typologies (section 2). At each point we recorded a 20-30 s reading (with a 10 s audio clip) in ODK Collect, submitted live to a KoboToolbox server with automatic timestamp and GPS. Phones were held at 1.2 m (SPB positioning) at varied distances from the road, by three cross-calibrated collectors, from 05:00 to 23:00 with emphasis on rush hours. Weather (Open-Meteo) and spatial context (OSM) are added.

3.2 Reproduction of the Sunbird pipeline

We reproduced the full Sunbird chain on their Uganda dataset (notebooks 01-06): cleaning, audio QC, morphology features, figures, surrogate model. Our statistics match the paper (median 49 dB vs their 45-50 dB).

3.3 Model features

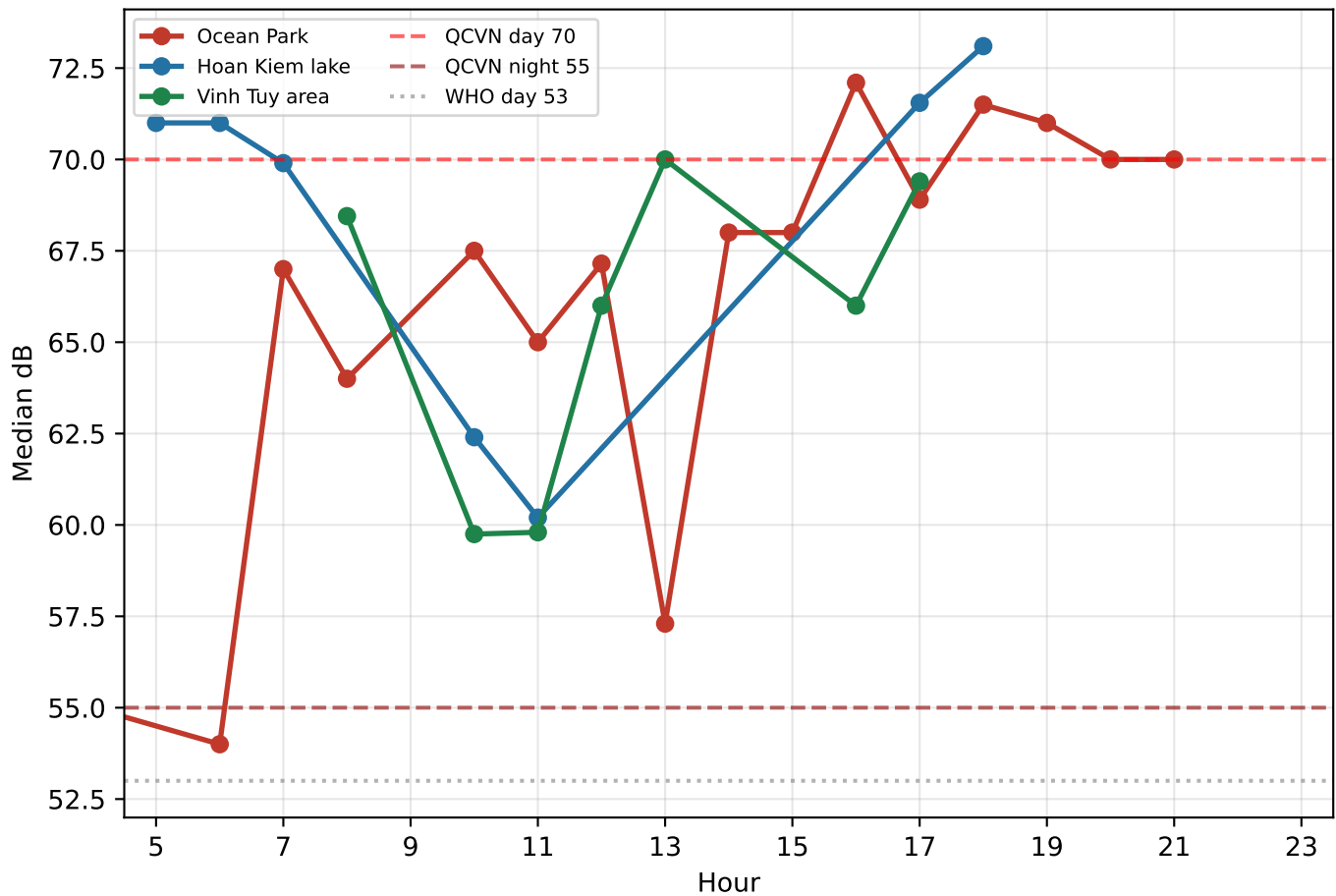
Morphology within 300 m of each point (OpenStreetMap): built-area ratio, road density, intersection count, distance to nearest road; plus hour of day and weekend flag. Model: LightGBM.

3.4 Standards used

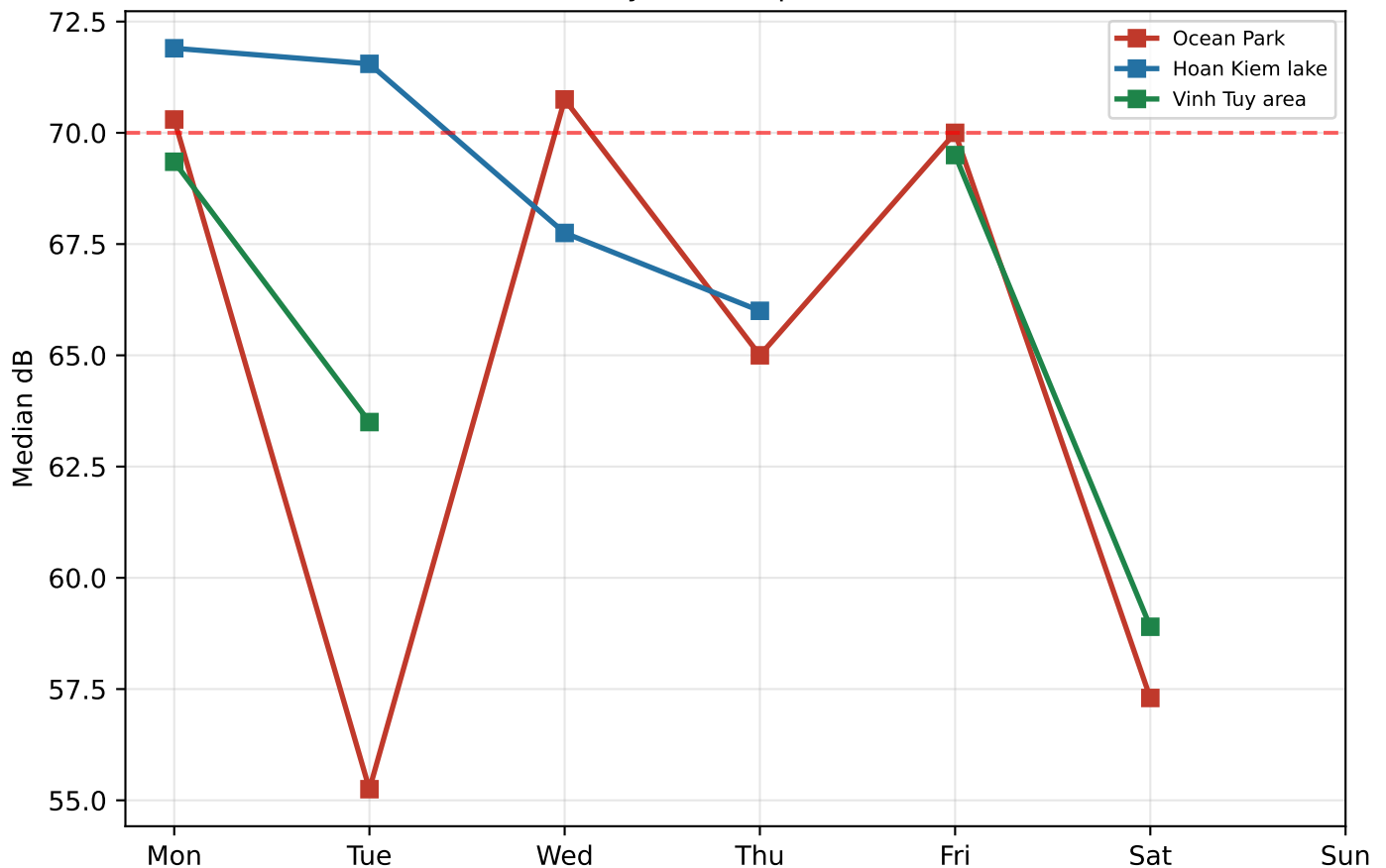
Standard	Day (6-21h)	Night (21-6h)
QCVN 26:2010/BTNMT (Vietnam, ordinary area)	70 dB	55 dB
WHO (road-traffic guideline)	53 dB	45 dB

4. Data analysis: temporal patterns

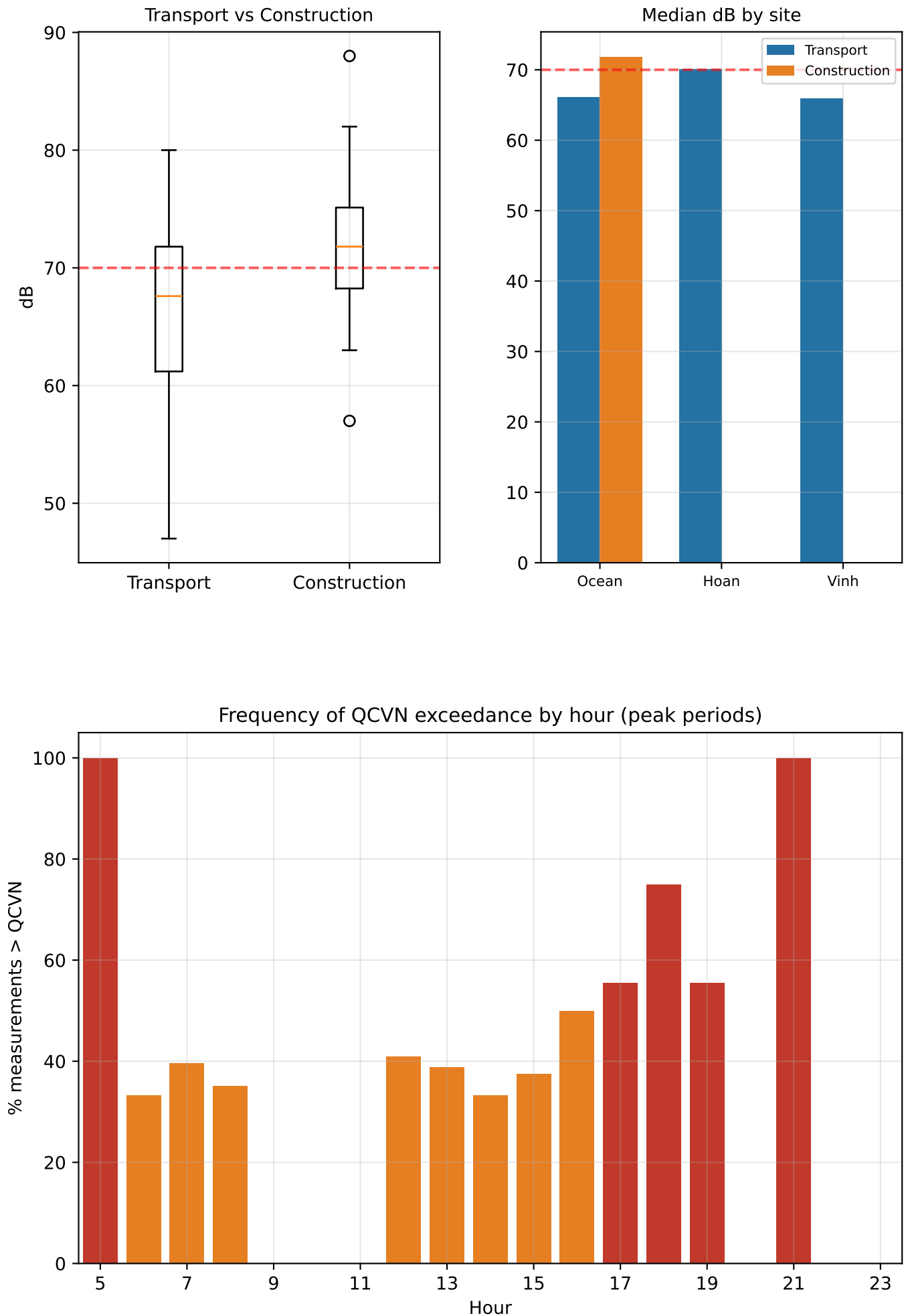
Hourly pattern within a day (capture window 05:00-23:00)



Day-of-week pattern



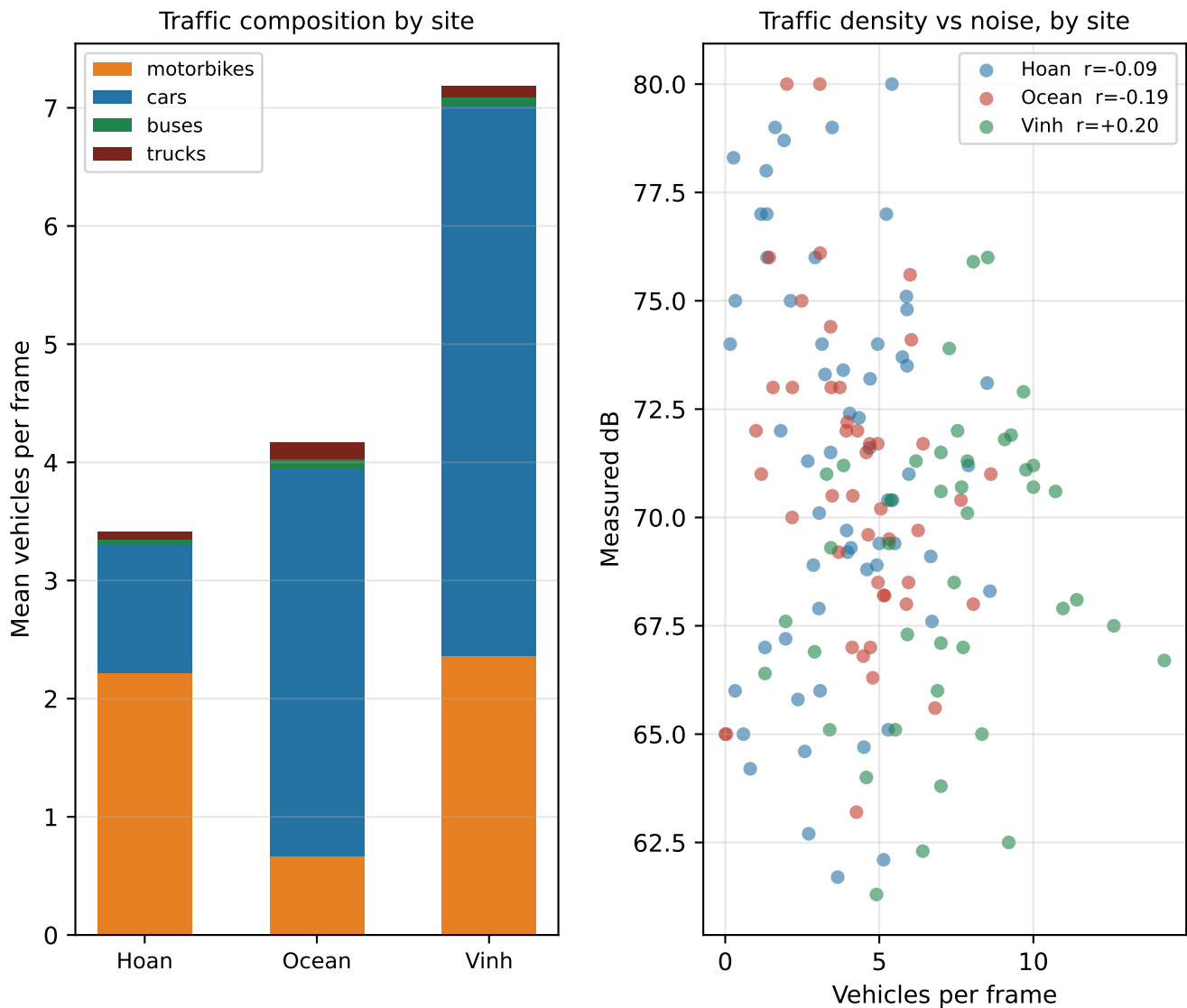
4. Data analysis: sources & exceedances



4. Data analysis: traffic videos & weather

147 traffic videos (~25 s each) matched to their measurement by timestamp (median gap 15 s).

Vehicles counted with YOLOv8 at ~1 frame/s (3726 frames analysed): average vehicles visible per frame.



Findings: traffic composition matches the zone typologies: Hoan Kiem is motorbike-dominated (65%), Ocean Park car-dominated (84%), Vinh Tuy the densest corridor (up to 14 vehicles/frame). The link between density and noise is site- and congestion-dependent, not a simple rule: in most sessions more vehicles means more noise, but the two most congested Ocean Park sessions (7-8 vehicles/frame, near-gridlock) show the opposite (r about -0.35) - slow, jammed traffic can be quieter than a faster, sparser flow. Note: small motorbikes are harder to detect than cars and parked vehicles are counted too (no motion filter yet), so motorbike shares are a lower bound.

Weather: no robust effect. Raw correlations (temperature +0.26, rain -0.29) vanish once hour of day and session are controlled: quiet-point campaigns happened to fall on rainy days. Wind shows no effect on readings (no microphone artefact).

5. Predictive model

LightGBM mapping urban morphology + time of day to noise level (dB).

5.1 Key finding: transfer fails, direct training works

The Uganda-pretrained model transfers poorly to Hanoi even after offset calibration ($R^2 < 0$): the noise-morphology relationship learned in Kampala does not hold here, as in the Barcelona cross-city experiment. Training directly on our own measurements works, and is our method.

Method	r	R ²	MAE (dB)
Direct training, spatial CV (honest)	0.69	0.45	4.2
Direct training, random split (optimistic)	0.70	0.48	4.1
Uganda to Hanoi transfer (comparison)	0.07	−1.26	7.9
Barcelona reference (for context)	0.66	0.61	n/a

Spatial CV: tested on locations the model never saw in training (the honest number). Random split: test points can sit metres from training points, which inflates scores; kept as an optimistic upper bound.

5.2 Per-site generalization (leave-one-site-out)

Test site	R ²	Comment
Ocean Park	0.21	best case (largest, most varied sample)
Vinh Tuy	−0.68	distinct transport-corridor typology
Hoan Kiem	−0.37	distinct old-quarter morphology

A stress test: each site is predicted by a model trained on the other two only. Per-site R^2 on such small samples is noisy; the spatial CV above is the reliable number.

Reading the metrics: r = does the model rank places correctly; R^2 = are predicted dB values accurate. Direct training delivers both: r 0.69, R^2 0.45.

6. Agent-based simulation (GAMA)

The predicted noise map is loaded into an agent-based model with moving vehicles, construction sites and interactive scenario controls.

6.1 What the simulation contains

Layer	Source	Status
Background noise level, per hour	LightGBM on our 363 measurements	predicted
Traffic density and vehicle mix	147 videos, YOLO counts	measured
Construction excess (+2 dB at 56 m)	our 32 near-site measurements	calibrated
Vehicles as sound sources	not identifiable in our data	excluded
Traffic-volume scenario	10 log10 of the volume factor	physical law

6.2 Scenario controls

Hour of day (5h-21h, each hour predicted separately, vehicle numbers follow the traffic measured at that hour) · traffic volume (x0.2 to x3) · construction on/off with working hours · mitigation: 30 km/h zone (-3 dB) or pedestrianisation (traffic x0.2).

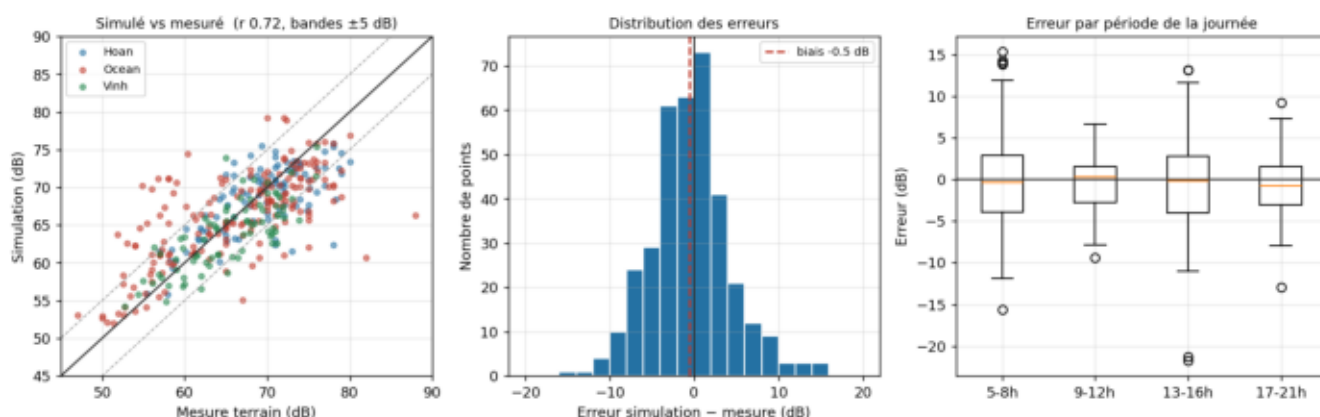
Calibration note: we tried to derive a per-vehicle emission level from our own data (non-negative energy regression on the 147 matched videos). All three coefficients came out at zero - at a given site, the number of visible vehicles does not explain the measured level (R2 0.008-0.042). Rather than inject invented values, vehicles are displayed as a calibrated picture of the fleet but carry no sound in the computation.

6.3 Does the simulation reproduce our measurements?

Metric	Value
Measurements compared	360
Bias (simulated - measured)	-0.52 dB
Mean absolute error	3.68 dB
Correlation r	0.72
Within +/- 5 dB	74 %

Each field measurement is compared with the grid cell it falls in (median 16 m away), at the hour it was taken.

Caveat: this is an in-sample check - the model that produces the grid was trained on these same points. It measures the fidelity of the chain model -> 40 m grid -> GAMA, not generalisation. The honest generalisation figure remains the spatial cross-validation of section 5: R2 0.45 / MAE 4.2 dB.



6. Limitations

- The model interpolates well within the sampled areas but extrapolates poorly to a district it has never seen: city-wide prediction would need more urban typologies covered.
- Instantaneous readings carry ± 5 dB of irreducible noise (a passing bus), capping the achievable R^2 near 0.6; repeated measurements at fixed points would raise it.
- Consumer sound meters: a constant calibration bias is correctable, clipping above 90 dB and wind noise are not. Weekend coverage is still light compared to weekdays.

7. Next steps

- Add weekend sessions and repeated measurements at fixed locations to average out instantaneous variation.
- Vehicle counting from the 83 traffic videos via computer vision, each matched to its measurement point (transport composition as a model feature).
- GAMA simulation: import the noise map, add a traffic-volume slider and what-if scenarios (pedestrianisation, peak-hour traffic).
- Manuscript: methods (Sunbird reproduction + transferability study), results, discussion.

8. Summary

363 calibrated measurements across 3 districts confirm high exposure (39% exceed QCVN). A model trained on our data reaches R^2 0.45 / r 0.69 / MAE 4.2 dB (spatial CV), solid for instantaneous smartphone data. Cross-city transfer from Uganda fails: a documented, useful methodological result.